

SkillPath

AI-Aware Developer Career Recommendation System

Recommends developer job roles from a person’s skills, experience and AI usage, and explains each one with evidence on AI exposure, typical pay and skill gaps.

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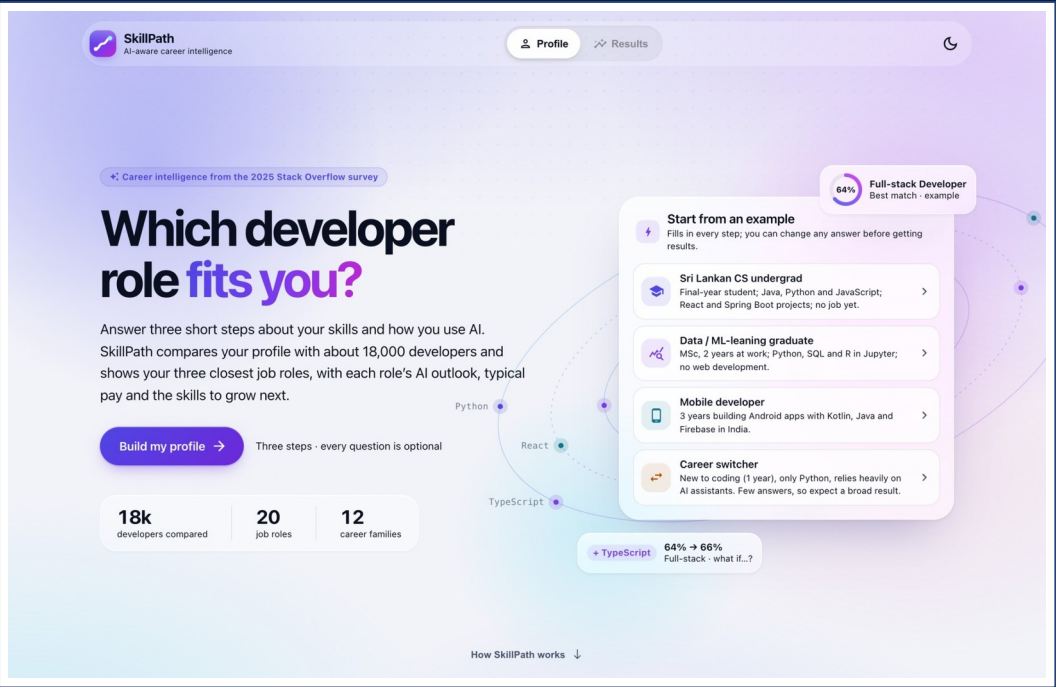
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The SkillPath web application (start page)

49,123

survey responses

20 / 12

job roles / families

479

model features

83.9%

top-3 role accuracy

Problem Statement and Motivation

WHY CHOOSING A PATH IS HARD

1

Many specialised roles

Back-end, front-end, mobile, data, ML, DevOps, cloud, security, QA: each has its own technology stack.

2

Guidance is generic

Articles, role descriptions and job adverts are not matched to a person's actual skill profile.

3

AI is changing every role

Students ask how much of a role's everyday work AI already does. Advice that ignores this is incomplete.

WHAT A USEFUL ANSWER MUST SHOW

AI

AI exposure

The share of a role's everyday tasks already done with AI, and whether people in the role see AI as a threat.

\$

Salary evidence

What people in the role earn in comparable circumstances: same country or region, same experience band.

+

Skill gaps

The specific technologies that separate a person's current skills from a target role.

THE DATA MINING PROBLEM

Given a person's self-reported technologies, experience, education, location and AI usage, **predict which developer job roles their profile most resembles**, and support each suggestion with evidence on AI exposure, salary and skill gaps from the same survey.

Aim, Objectives and Scope

AIM Design, build and evaluate an AI-aware developer job role and career path recommendation system by mining the Stack Overflow Developer Survey 2025.

SIX OBJECTIVES

- 1 Understand the data**
Explore the survey, assess its quality, find the relevant attributes
- 2 Prepare it without leakage**
A reproducible pipeline that is reused unchanged at prediction time
- 3 Compare algorithms**
At least four classifiers, with validation and metrics fit for imbalance
- 4 Optimise, select, evaluate**
A selection rule fixed in advance; one evaluation on held-out data
- 5 Derive role insights**
AI Exposure Index, perceived AI threat, salary benchmarks, skill gap
- 6 Deploy and test**
A web API and an easy-to-use web app, verified by systematic testing

SCOPE AND INTENDED USE

20

skill-defined job roles

12

role families

TARGET USERS

Students and early-career developers

IN SCOPE

- Top-3 roles and families, with probabilities
- AI exposure, AI threat, salary and skill gap per role
- Web API and web app for a single user
- Unit, integration and system testing

OUT OF SCOPE

- Management, executive and non-technical roles
- Age, gender or other sensitive attributes as inputs
- Hiring, screening or salary-setting decisions
- Storing or logging user input; large-scale hosting

Intended use: suggestions for learning and career planning. A low score means a profile is unlike that role's survey respondents, not that the person cannot do the job.

Dataset and Exploratory Data Analysis

Stack Overflow Annual Developer Survey 2025

49,123

responses

170

attributes

ATTRIBUTES USED

Job role: class label (20 roles kept)

Technologies: 7 areas, used and wanted

Experience, education: years coding / working

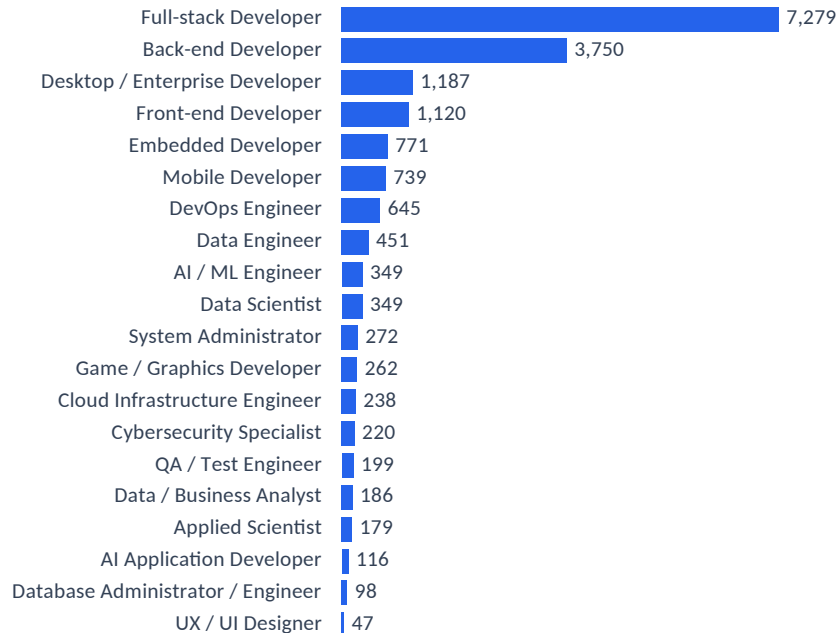
Location: 177 countries → 13 regions

AI usage: attitudes + 13-task matrix

Pay: benchmark only, never an input

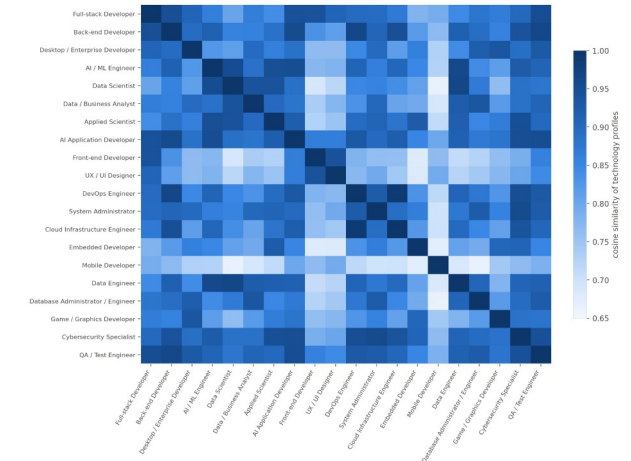
Licence: ODbL v1.0. The app shows the attribution on every result.

Class imbalance: 20 job roles, training split



Full-stack is 39.4% of 18,457; UX / UI Designer has 47. Ratio about **155:1**

Similarity of role technology profiles



Cosine similarity: **0.99** Cloud Infrastructure vs DevOps; **0.97** Data Engineer vs Data Scientist

No algorithm can fully separate roles that use the same tools.

FINDING 155:1 imbalance

→ Macro-F1 as the primary metric; stratified split and folds

FINDING Near-identical roles

→ Recommend three roles plus family totals, not one answer

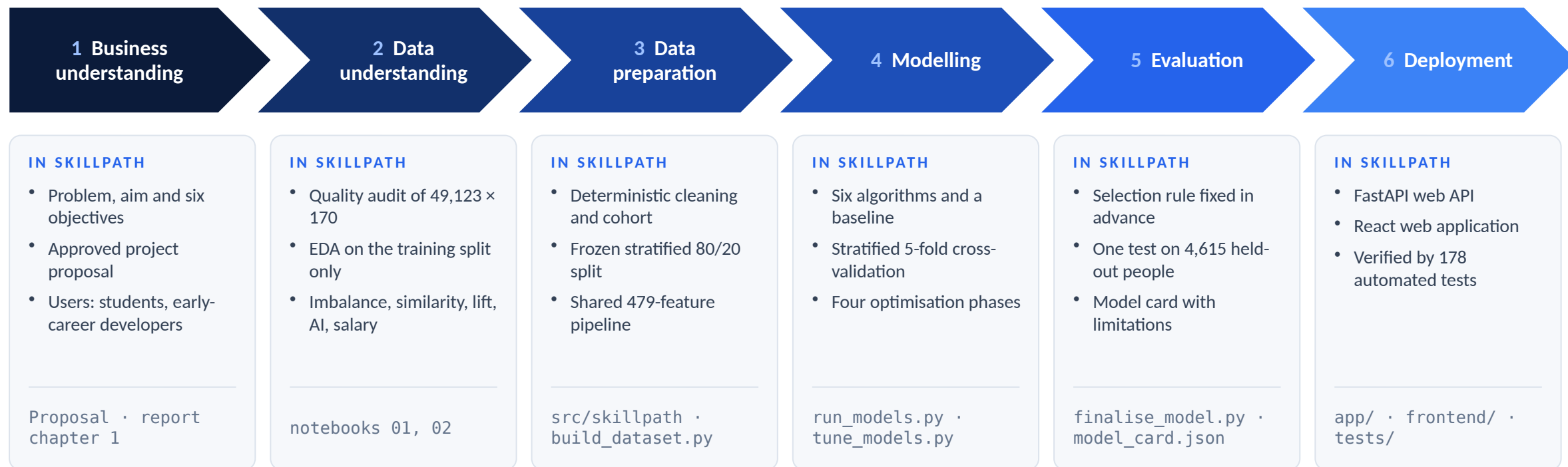
FINDING Distinctive technologies

→ Swift is 9.7× as common in Mobile: multi-hot flags, lift for skill gap

FINDING AI use differs by role

→ 19.1 of 100 tasks now, 39.0 expected; 14.4% feel threatened

CRISP-DM Methodology



Iterative, not a waterfall: modelling fed back into preparation: scaling binary flags cost 0.025 macro-F1, and class weighting was dropped for Logistic Regression.

One shared package

Notebooks, scripts and the web backend use the same skillpath code

Split first, learn later

Only fixed rules run before the split; the rest is fitted on training rows

Preprocessing inside the model

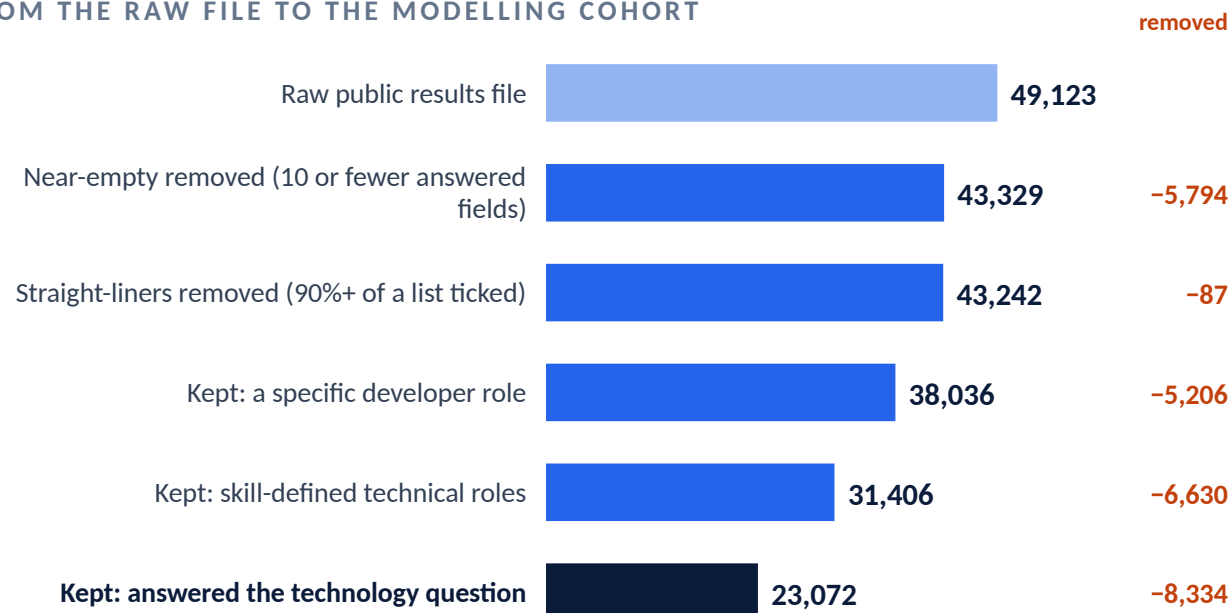
Refitted on every fold; the API applies the identical transformation

Test split opened once

Read by one script, after the model is chosen; seed 42 throughout

Data Cleaning and Preprocessing

FROM THE RAW FILE TO THE MODELLING COHORT



STRATIFIED 80 / 20 SPLIT OF THE 23,072 (BEFORE ANYTHING IS LEARNED)

Train **18,457** cross-validation, tuning, insight tables

Test **4,615**

Stratified on JobRole, seed 42, response IDs frozen. Largest class-share gap between the two sides: 0.02 points. The smallest role keeps 12 test examples.

Deterministic cleaning

BEFORE THE SPLIT

- Rules use questionnaire facts (38 of 42 languages), not data statistics, so they cannot leak
- Blank work experience → 0 when years coding is given (2,919 rows, per the survey's instruction)
- Experience impossible for the age band → missing (73 + 100 values)

Missing values and encoding

FITTED ON TRAINING ROWS

- Technology lists: 0/1 flags plus an "unknown" indicator; technologies are never imputed
- Years and ordinal scales: training median plus a missing-indicator column
- Rare technologies (< 0.5%) pooled into OTHER; 177 countries → 13 regions

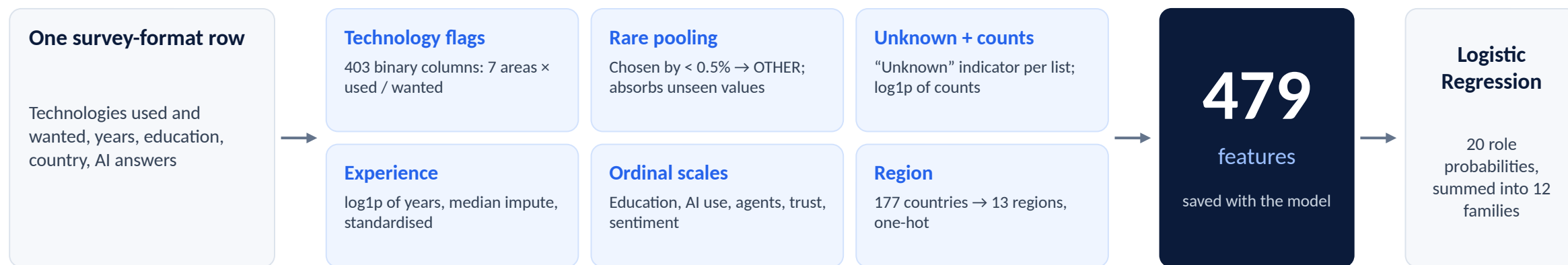
Leakage prevention

CHECKED BY UNIT TESTS

- Target, pay source columns and Age are never model inputs
- The preprocessor sits inside the model, so every cross-validation fold refits it
- Salary, AI and skill tables exclude the 4,615 test respondents

Feature Engineering and Supporting Insights

ONE SCIKIT-LEARN PIPELINE, FITTED ON TRAINING ROWS AND REUSED UNCHANGED BY THE WEB API



"None" is not "not answered"

Empty list after answering "No" to the area
→ true zero: every flag 0, unknown = 0

Empty list because the question was skipped
→ unknown = 1, technologies never imputed

SUPPORTING INSIGHTS SHOWN WITH EACH ROLE (NEVER MODEL INPUTS)

AI Exposure Index

- $100 \times$ mean over 13 development tasks: mostly with AI = 1, partly = 0.5
- All roles: 19.1 now, 39.0 with planned use; 14.4% see AI as a threat

Salary peer-group benchmark

- Unit errors removed with a robust z-score of log pay within country ($|z| > 3.5$)
- Median and middle half for the most specific peer group of 30 or more people

Lift-based skill gap

- Lift = share of the role using a technology $\div$ share across all roles
- Distinctive: 20%+ of the role and lift ≥ 1.1 ; shows up to five not yet used

Classification Algorithms and Validation

Supervised multi-class classification

20	479	18,457	155:1
job-role classes	features	training rows	class imbalance

SIX ALGORITHMS AND A BASELINE

- **Hist Gradient Boosting** boosted trees; each round corrects earlier errors
- **Logistic Regression** linear boundaries; good probabilities
- **Random Forest** bagged trees; captures skill combinations
- **Linear SVM (calibrated)** maximum margin; calibrated for probabilities
- **Complement Naive Bayes** assumes independent features
- **k-Nearest Neighbours** cosine similarity: “similar skills, similar job”
- **Dummy (most frequent)** baseline: always answers Full-stack

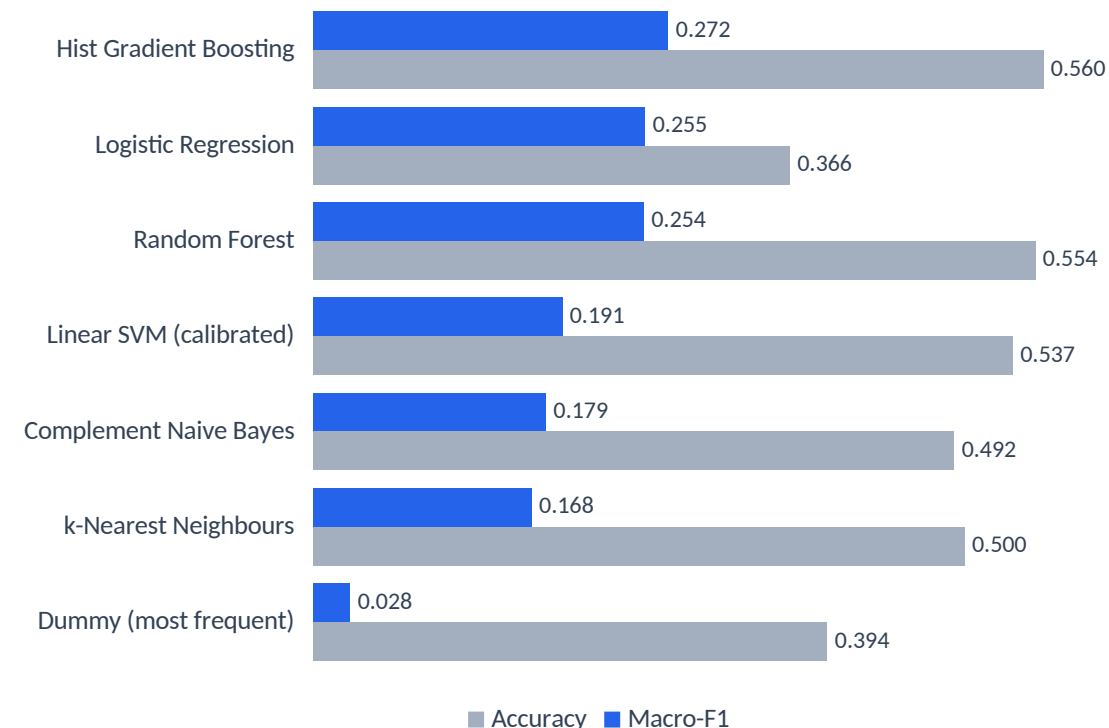
Stratified 5-fold cross-validation

Same folds for every run (seed 42). The preprocessor is refitted in each fold. About 9 examples of the smallest role per validation fold.

Macro-F1 is the primary metric

The plain mean of the 20 per-role F1 scores: every role counts equally, so ignoring rare roles is penalised.

Stage 6 results: macro-F1 against accuracy (5-fold mean)



Accuracy hides what macro-F1 shows. The Dummy model reaches 39.4% accuracy with no skill at all, yet its macro-F1 is 0.028. The top three models are within 0.019 of each other.

Model Optimisation and Final Model Selection

A Imbalance handling

None, class weights or oversampling?

- Weighting harms Logistic Regression: 0.2535 $\rightarrow$ 0.3003 without it
- It helps Random Forest: 0.1869 $\rightarrow$ 0.2548

B Feature set and selection

Which inputs earn their place?

- “Want to learn” lists matter: -0.009 without them
- Feature selection does not help; scaling binary flags costs 0.025

C Hyper-parameter search

Randomised search, same folds

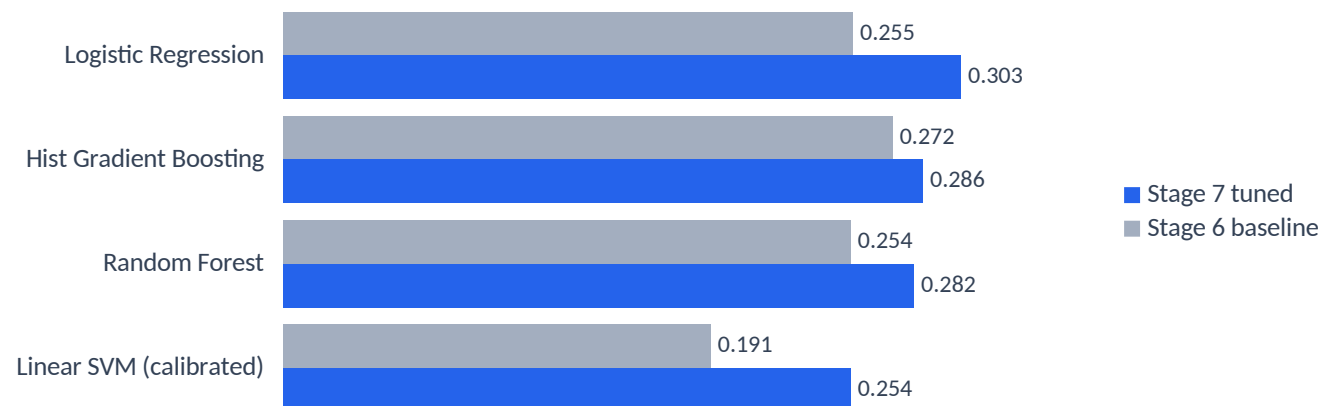
- Stronger regularisation wins: C = 0.218
- The search was free to pick class weighting and rejected it

D Tuned against baseline

Did the ranking survive tuning?

- All four tuned models improve
- Logistic Regression overtakes Gradient Boosting

Macro-F1 before and after optimisation (5-fold mean, training split)



Why class weighting hurt Logistic Regression

Re-weighting inflates rare-role probabilities everywhere: a little recall gained, far more precision lost. Removing it adds +0.047 macro-F1.

Tuning changed the winner

Gradient Boosting led Stage 6 (0.2724). After tuning, Logistic Regression leads 0.3030 to 0.2860 and fits in 3.3 s.

SELECTION RULE, WRITTEN BEFORE THE RESULTS WERE READ

Highest 5-fold macro-F1 on the training split. Models within one standard deviation of the best are tied, and the cheapest to fit wins.

FINAL MODEL

Logistic Regression

C = 0.218, no class weighting; the only model within one SD of the best

Final Model Evaluation

Logistic Regression, refitted on 18,457 training rows and evaluated **once** on 4,615 held-out respondents

83.92%

Top-3 job-role accuracy: the true role is among the three shown

87.78%

Career-family top-3 accuracy

54.93%

Top-1 accuracy

0.2845

Macro-F1 (20 roles)

0.4149

Family macro-F1 (12 families)

What this means in practice

- For about 84 of every 100 held-out people, their real role is on the screen
- Roughly double the majority-class baseline (top-3 of 0.41 in cross-validation)
- The right career family is in the top three 87.8% of the time

Honest validation: cross-validated estimate against the test result

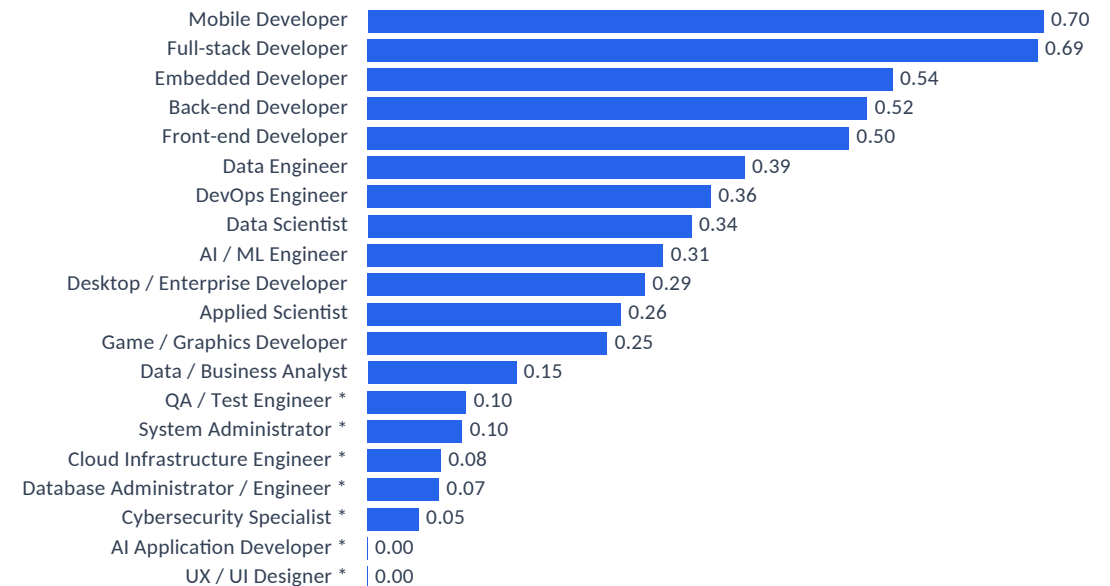
	5-fold CV (train)	Held-out test	Gap
Macro-F1	0.3030	0.2845	-0.0185
Top-3 accuracy	0.8387	0.8392	+0.0005
Log loss	1.4146	1.4146	0.0000

Limits we report, not hide

- Top-1 is modest: Full-stack (39% of the data) absorbs uncertain profiles
- 7 of 20 roles have recall below 0.10 and carry a “low confidence” badge; two are never predicted first

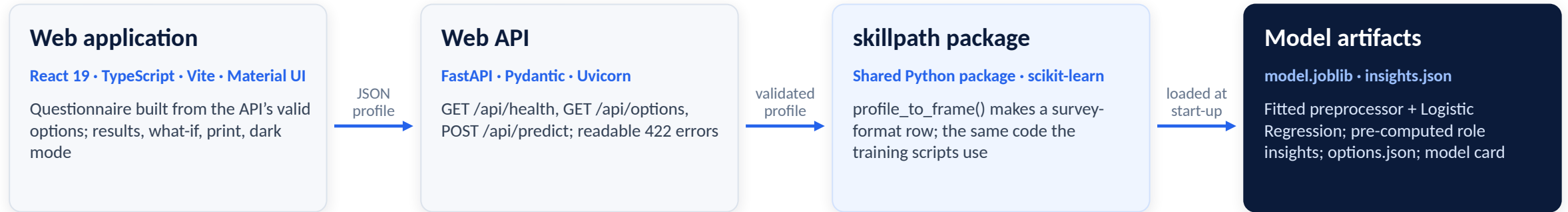
F1 per job role on the held-out test set

* low confidence (recall < 0.10)



System Architecture and Technology Stack

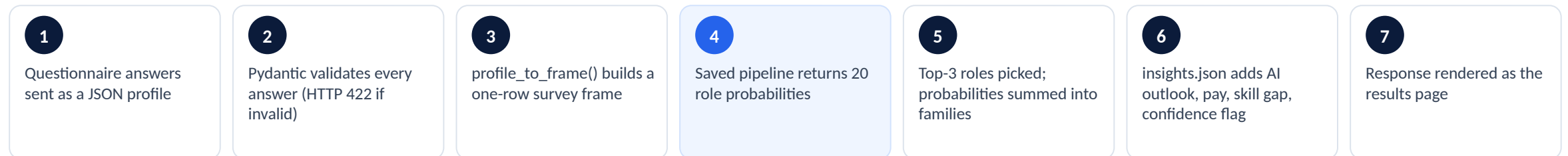
ONLINE: ONE RECOMMENDATION



OFFLINE: THE DATA MINING WORKFLOW THAT PRODUCES THE ARTIFACTS



PREDICTION FLOW OF POST /API/PREDICT



No preprocessing is re-implemented in the backend • nothing the user sends is stored or logged • about 15 ms per prediction, including insights

Application Demonstration and Testing

Programming languages

Programming, scripting and markup languages

✓ 5 used · 2 wanted

Used for extensive work in the past year

Java Python JavaScript

HTML/CSS SQL

Popular: + Bash/Shell (all shells) + C# + Rust

+ C++ + PowerShell

Want to work with next year

TypeScript Go

☐ I don't use any of these

Databases

Database environments and data stores

✓ 2 used · 1 wanted

Used for extensive work in the past year

MySQL MongoDB

Popular: + SQLite + Redis

+ Microsoft SQL Server + MariaDB

+ Elasticsearch

Want to work with next year

PostgreSQL

☐ I don't use any of these

Cloud and dev platforms

Cloud providers, containers, package managers and build tools

✓ 2 used · 2 wanted

Used for extensive work in the past year

npm Pip

Popular: + Kubernetes + Vite

+ Microsoft Azure + Homebrew

+ Google Cloud

Want to work with next year

Web frameworks

Web frameworks and web technologies

✓ 4 used · 1 wanted

Used for extensive work in the past year

React Node.js Express

Spring Boot

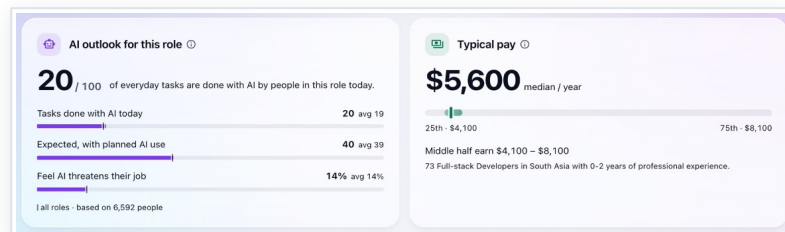
Popular: + ASP.NET Core + Vue.js + jQuery

+ Angular + FastAPI

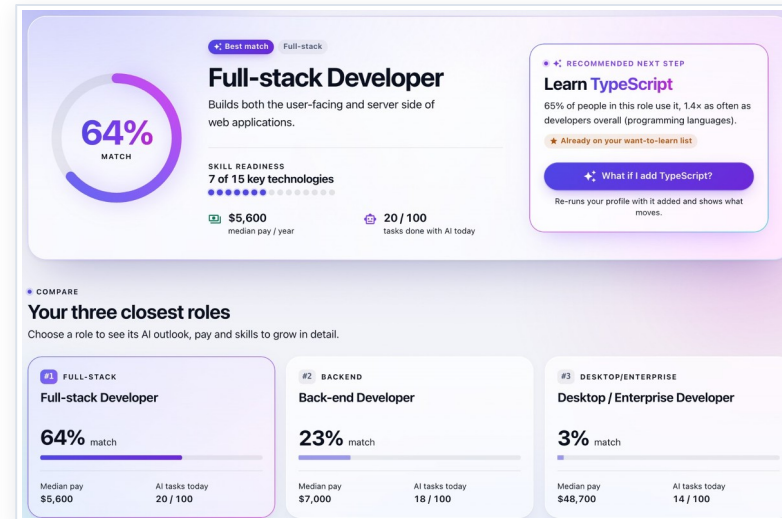
Want to work with next year

Next.js

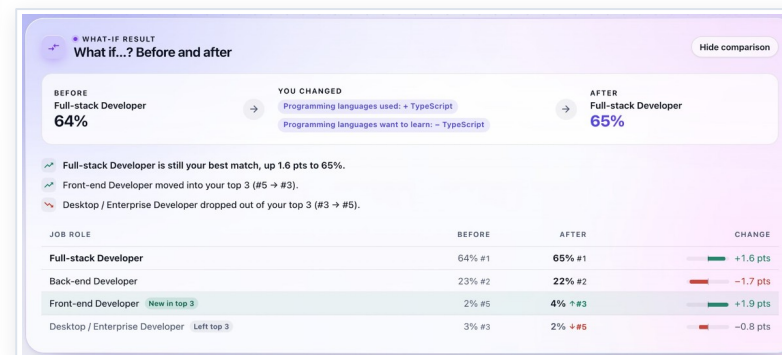
1 Questionnaire: technologies used, wanted, or "none"



3 Evidence per role: AI Exposure Index and salary peer group



2 Results: best match and the three closest roles



4 "What if...?" Adding a skill re-runs and compares

TESTING

178 automated tests

177 passed · 1 skipped · 0 failed

Unit: data and model
pytest 36

Unit: role insights
pytest 30

Integration: web API
pytest + TestClient 41

Unit: web app logic
Vitest 56

System: whole app
Puppeteer, real Chrome 15

Training-serving parity

Maximum probability difference of exactly 0.0 between the API path and the model (300 rows; repeated over HTTP with 150 held-out respondents).

- End-to-end browser test: 15 scenarios, including phone layout and dark mode
- Median 15.4 ms per prediction

Limitations, Future Work and Conclusion

Key achievements

- Leakage-free pipeline: 49,123 responses to 479 features
- Systematic comparison; tuning changed the winner to Logistic Regression
- 83.9% top-3 role and 87.8% top-3 family accuracy on held-out data
- Web API and web app, verified by 178 automated tests

Limitations

- Survey respondents are not a random sample of developers
- Job roles and skills are self-reported
- Weak on rare roles; two are never predicted first
- A 2025 snapshot; worldwide salary benchmarks may not match local pay

Future work

- Retrain annually on each new survey
- Better support for rare roles (more data, hierarchical models)
- Personalised explanations from model coefficients
- Learning resources and local-currency salaries
- Usability and accessibility testing with students

CONCLUSION

SkillPath applies the full data mining process, from data understanding to a tested deployment, to turn a developer's skills and AI usage into three explained, evidence-backed role recommendations, with an honest evaluation and transparent limits.

Thank You

Questions?

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SOURCE CODE

github.com/PasinduSuraweera/skillpath

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